Fill out this table ONLY with the names of those students **present** in your IN PERSON group meeting while you are completing the assignment.

Group Member names	Is the student present in this meeting?
Yash Rao	Yes
Edwin Huang	yes
Younwoo (Rachel) Park	yes
Kate Pun	yes
James Anderson	yes

What is the link to your Google doc?

https://docs.google.com/document/d/1sfsRXMSZMXM8JBnBb1WLEIttrQJJiuDRB_Wk9jwjjzM/edit?usp=sharing

A. Group Discussion

1. What are the key differences between GoBackN and Selective Repeat.

Go-Back-N and Selective Repeat differ mainly in retransmission and efficiency. Go-Back-N retransmits all frames after a lost or erroneous one, while Selective Repeat retransmits only the specific lost frames, making it more efficient. Go-Back-N's receiver discards out-of-order frames and has no buffering, whereas Selective Repeat buffers and reorders out-of-order frames, supporting more flexible and efficient data handling.

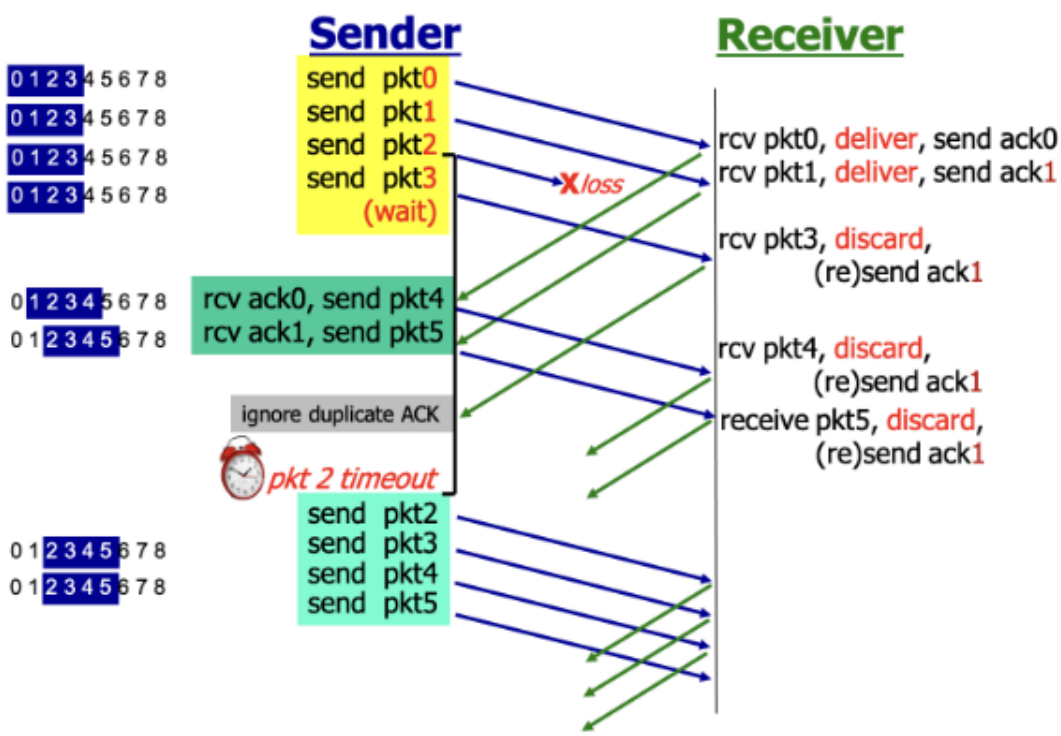
2. What similarities do you see with TCP's reliable data delivery mechanism? What aspect is like GBN and or Selective Repeat?

TCP's reliable data delivery mechanism has elements of both GBN and Selective Repeat. Like GBN, TCP uses acknowledgments, which can sometimes lead to retransmitting multiple segments if a loss is detected. However, TCP goes beyond GBN by incorporating features similar to Selective Repeat through the use of selective acknowledgments. This allows the sender to identify and retransmit only the specific missing segments, making the process more efficient. Essentially, TCP blends the straightforward acknowledgment system of GBN with the more focused retransmission strategy of Selective Repeat.

B. Answer the following questions:

1. Let's think about GoBackN:

Remember, "pkt0" is the first packet transmitted and it has SN=0. "pkt1" is the 2nd transmitted, and it has SN=1 etc. "send ack0" means an ACK is sent with the number 0 (indicating the highest in order received is SN 0).



Answer the following questions:

1. What does it mean that packet 1 is "delivered" ?

Packet 1 “delivered” means that packet 1 has been correctly received by the receiver process. The receiver has sent an acknowledgement, indicating successful delivery.

2. Why is packet 3 discarded?

Packet 3 was discarded because GoBackN requires packets to be received in sequence (Packet 3 was being received before packet 2). In other words, the sender waits for ACK 2 but never receives it because Packet 2 is lost in transit. So Packet 3 gets discarded due to the sender not receiving Packet 2.

3. When does the transmission window SLIDE over in this example?

The transmission window slides over when new acks are received that confirm that packets were delivered and received in order. In this example, the window slides over after the sender received acks for packet 0 and packet 1.

4. What is the benefit to the throughput when the window slides over?

The benefit is that it allows the sender to send new packets without waiting for an ACK for each individual packet. This increases the throughput because it allows the sender to continuously transmit packets in the window limit.

5. What are the Sequence Numbers of the new packets that are transmitted when the window slides?

Sequence numbers of the new packets would be 4, 5. (packets of sequence number 2 and 3 are also sent along, but they were also sent in the previous window).

6. After the timeout, why does the sender transmit packets with Sequence Numbers 2,3,4 and 5?

Sender packets with sequence number 2, 3, 4, 5 to recover the packet loss of packet 2. Since the packets need to be delivered in order, the sender resends all packets from packet 2 onward to ensure the receiver gets all missing packets and can resume in-order delivery.

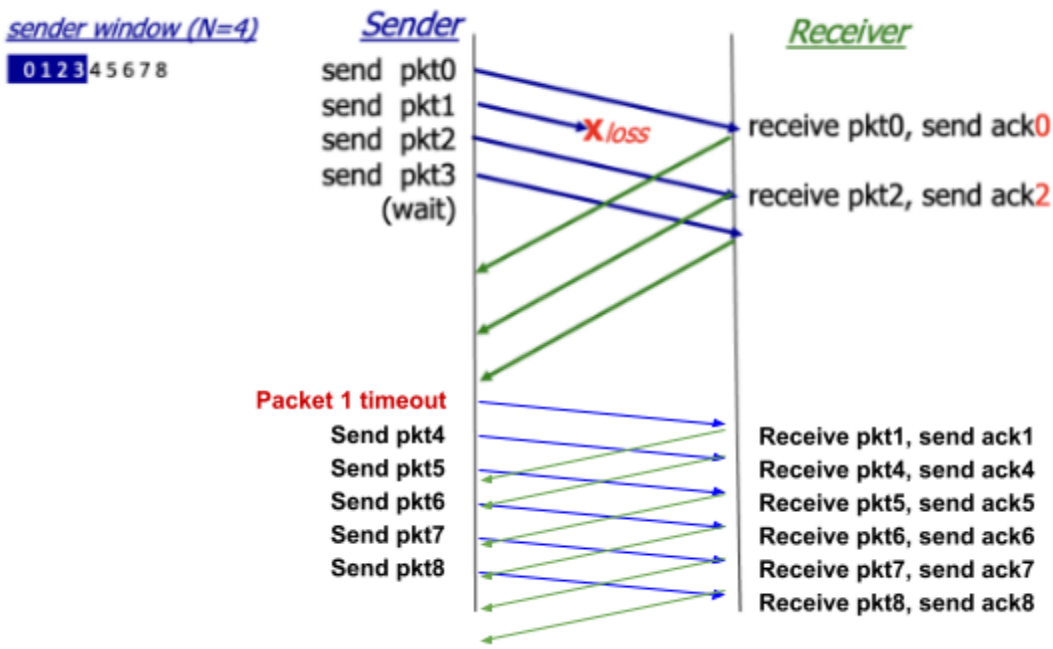
7. Once the ACKs are received for the retransmitted packets, what is the Sequence Number of the very next packet that the Sender can transmit?

Sequence number 6 because the very last packet sent had a sequence number of 5.

2. Let's think about Selective Repeat:

Complete the timing diagram to show the transfer of packets using Selective Repeat when pkt1 (the 2nd packet transmitted) is "lost". The drawing should continue until all packets are received at the Receiver. Don't forget the retransmission timer! Mark the SN and ACK# of all packets. When finished, what do you observe as the main difference from GoBackN?

The main difference from Go-Back-N is that in Selective Repeat, only the lost pkt1 is retransmitted, and subsequent packets (pkt2 and pkt3) are accepted and acknowledged even before pkt1 is received. This avoids redundant retransmissions and improves efficiency compared to GoBackN, where the sender would have to retransmit all packets starting from the lost packet.



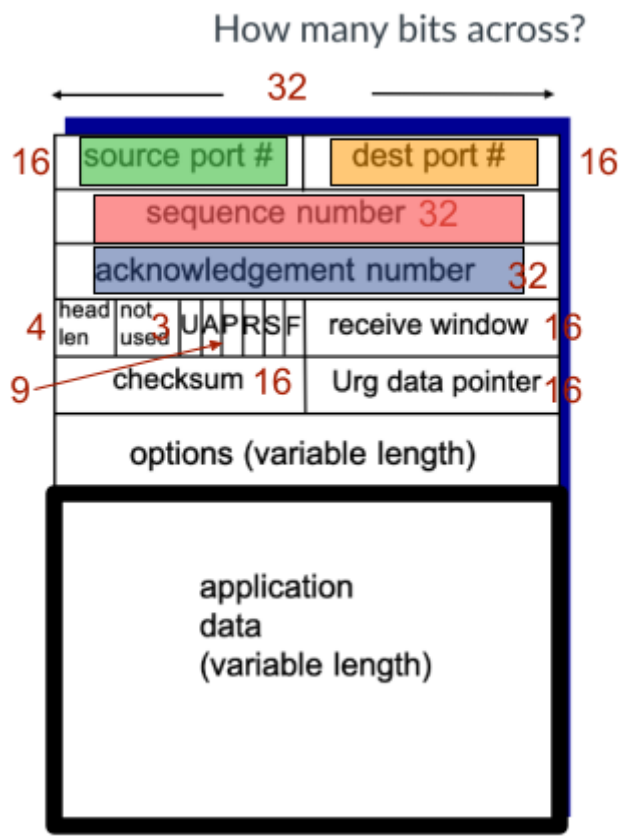
3. TCP ACKs

1. How does the TCP ACK differ from the GoBackN ACK?

TCP ACK differs from the GoBackN ACK because in TCP, the ACK acknowledges the next expected sequence number. If it receives packet 0 and 2 but not 1, it will still send ACKS for packet 0 and continue to do so until packet 1 is received. GoBackN ACK differs because it requires the sender to retransmit all the unacknowledged packets after a loss even if some of the packets have been received correctly.

2. What changes would need to be made to the TCP ACK so that the Sender would know exactly what byte might have been dropped in the network?
The TCP ACK would need to have information on the packet's size and which packet would've lost some data. It would also need the sequence number of the packet in question and the bitmap representing the bytes in place to help determine which byte was dropped in the packet.

4. TCP Header



Everyone marks up the questions below on the TCP header (shown above):

- Write the number of bits across each row of the header
  - Flags (UAPRSF): ***9 bits***
- Highlight the Sequence number in red
  - ***32 bits***
- Highlight the ACK number in blue
  - ***32 bits***
- Write the #bits allocated for SN? Write the #bits allocated for ACK?
  - The #bits for SN is 32 and the ACK is 32.
- Highlight the source port in green
  - ***Source port = 16***
- Highlight the destination port in orange
  - ***Destination port = 16***
- Write the #bits allocated for source and dest ports

    - The #bits for SP is 16 and the DP is 16.
- Outline the area where the Application Layer goes goes in Black Highlighter
- As you think about the bit stream of a TCP packet arriving at a destination, what bits are encountered first -- the Destination Port number or the Acknowledgement Number?
    - The destination port is encountered first, followed by the sequence number and the ACK number.
    - The destination number is in front of the byte stream of the TCP header, and hence it will be encountered before the ack number.

How long did it take you to complete this group assignment?

- 1 hour and 45 minutes